

Generative modeling through diffusions

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Motivation

Generative modeling

Sampling new data by learning the data distribution.

Two main early approaches:

- ▶ **Likelihood-based:**

- ▶ e.g., Autoregressive¹, VAEs², Normalizing Flows³
- ▶ *Limitation:* Needs tractable likelihood or approximations

- ▶ **Implicit:**

- ▶ e.g., GANs⁴
- ▶ *Limitation:* Training instability, poor convergence

¹ Larochelle et al. (2011). “The Neural Autoregressive Distribution Estimator”. In: *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*. PMLR.

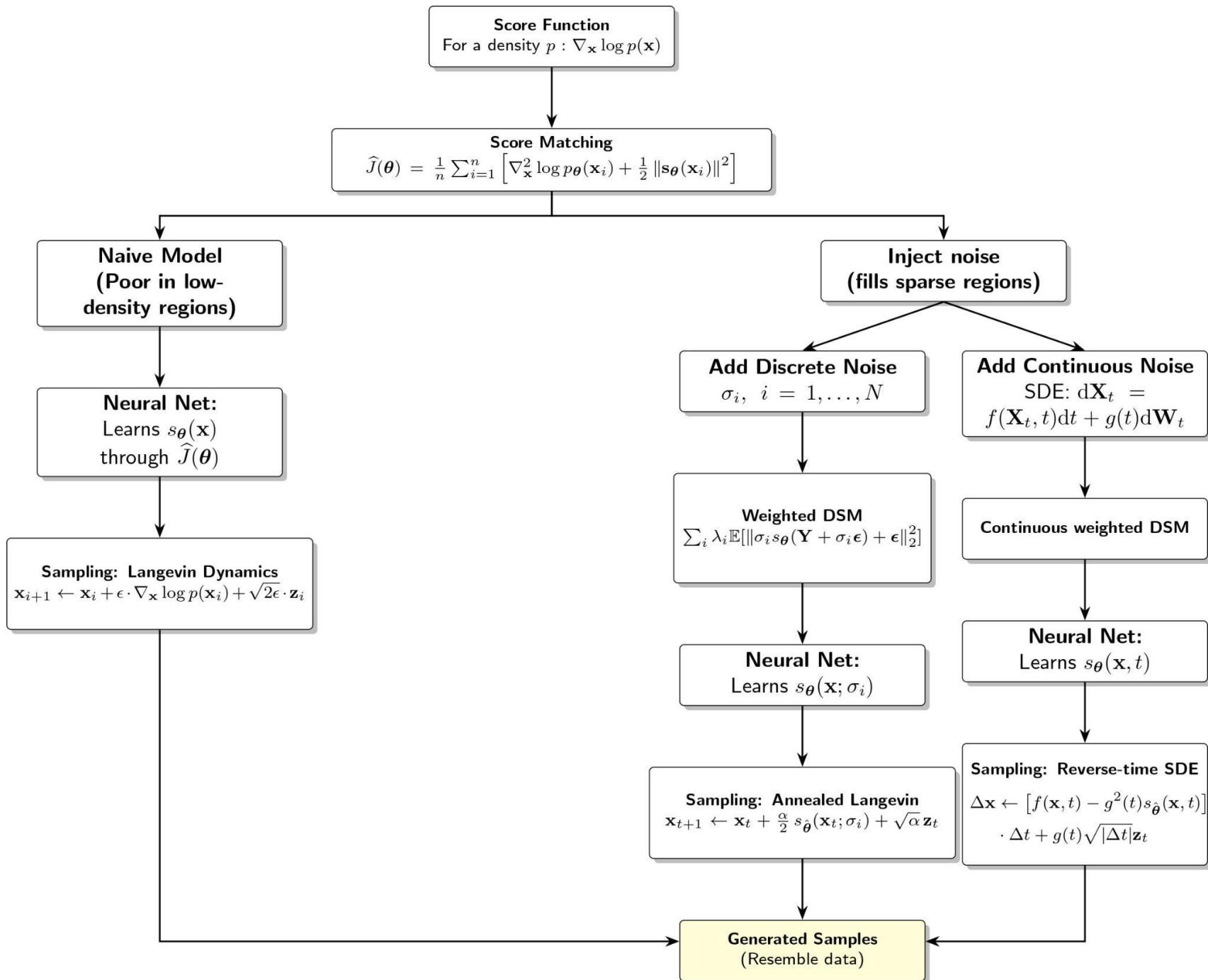
² Kingma et al. (2019). “An introduction to variational autoencoders”. *Foundations and Trends in Machine Learning*.

³ Papamakarios et al. (2021). “Normalizing flows for probabilistic modeling and inference”. *Journal of Machine Learning Research*.

⁴ Goodfellow et al. (2020). “Generative adversarial networks”. *Communications of the ACM*.

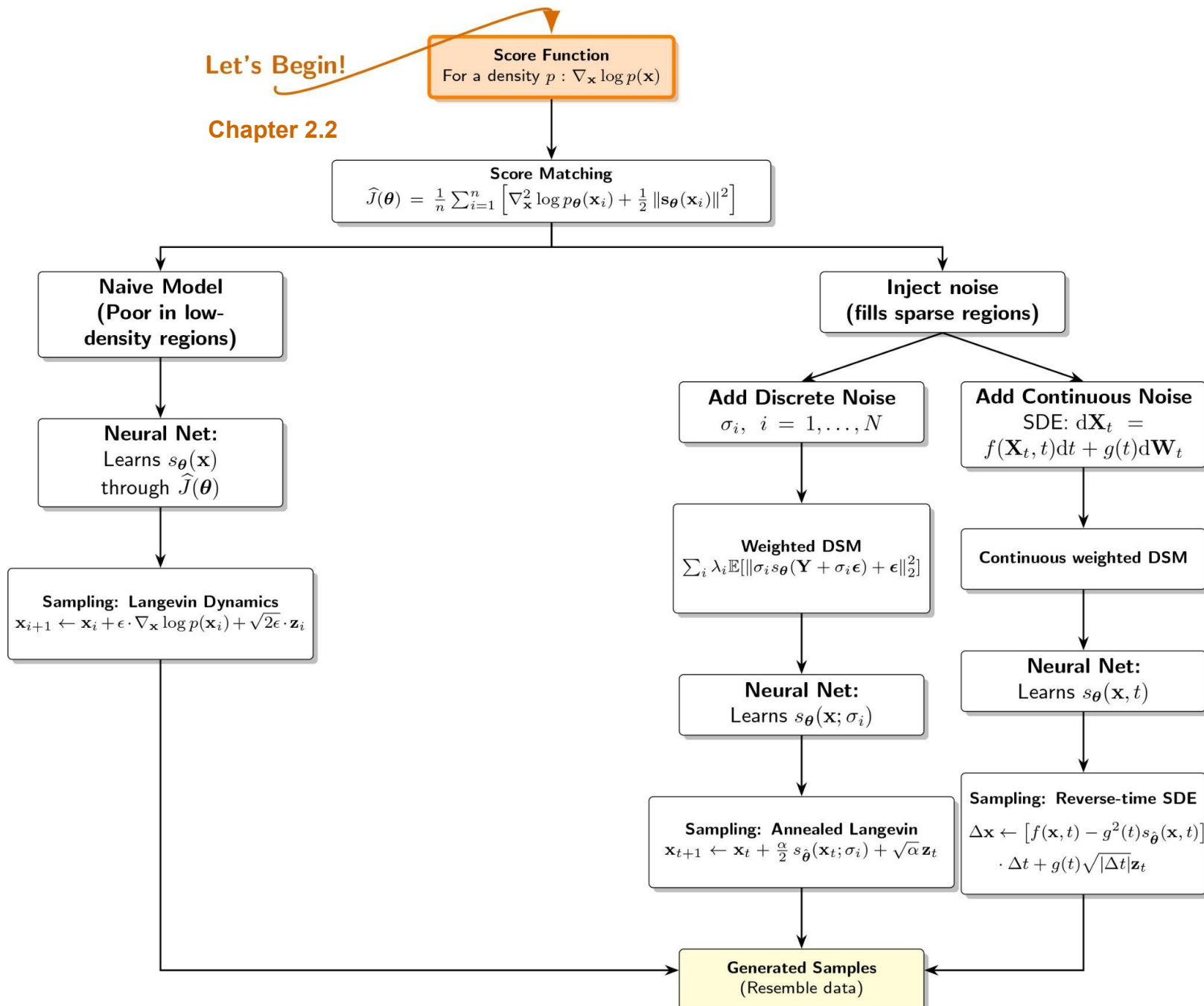
Main Goals

- ▶ Provide a **step-by-step explanation** of diffusion and score-based generative models.
- ▶ Clarify key concepts: **score function**, **score matching**, **Langevin dynamics**, **score SDEs**, **neural networks**.
- ▶ Illustrate how these concepts work together with **examples and visualizations**.
- ▶ Demonstrate training and sampling in with a **real dataset**.



Let's Begin!

Chapter 2.2



Score Function

- ▶ Defined as gradient of the true log density,

$$\mathbf{s}_0(\mathbf{x}) := \nabla_{\mathbf{x}} \log p_0(\mathbf{x}).$$

- ▶ A parametric model $\mathbf{s}_\theta$ tries to approximate the true score function,

$$\mathbf{s}_\theta(\mathbf{x}) \approx \mathbf{s}_0(\mathbf{x}).$$

- ▶ Taking the derivative with respect to $\mathbf{x}$ removes the need for normalizing constant,

$$\mathbf{s}_\theta(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}) = \nabla_{\mathbf{x}} \log \left(\frac{e^{-f_\theta(\mathbf{x})}}{\mathbf{Z}_\theta} \right) = -\nabla_{\mathbf{x}} f_\theta(\mathbf{x}).$$

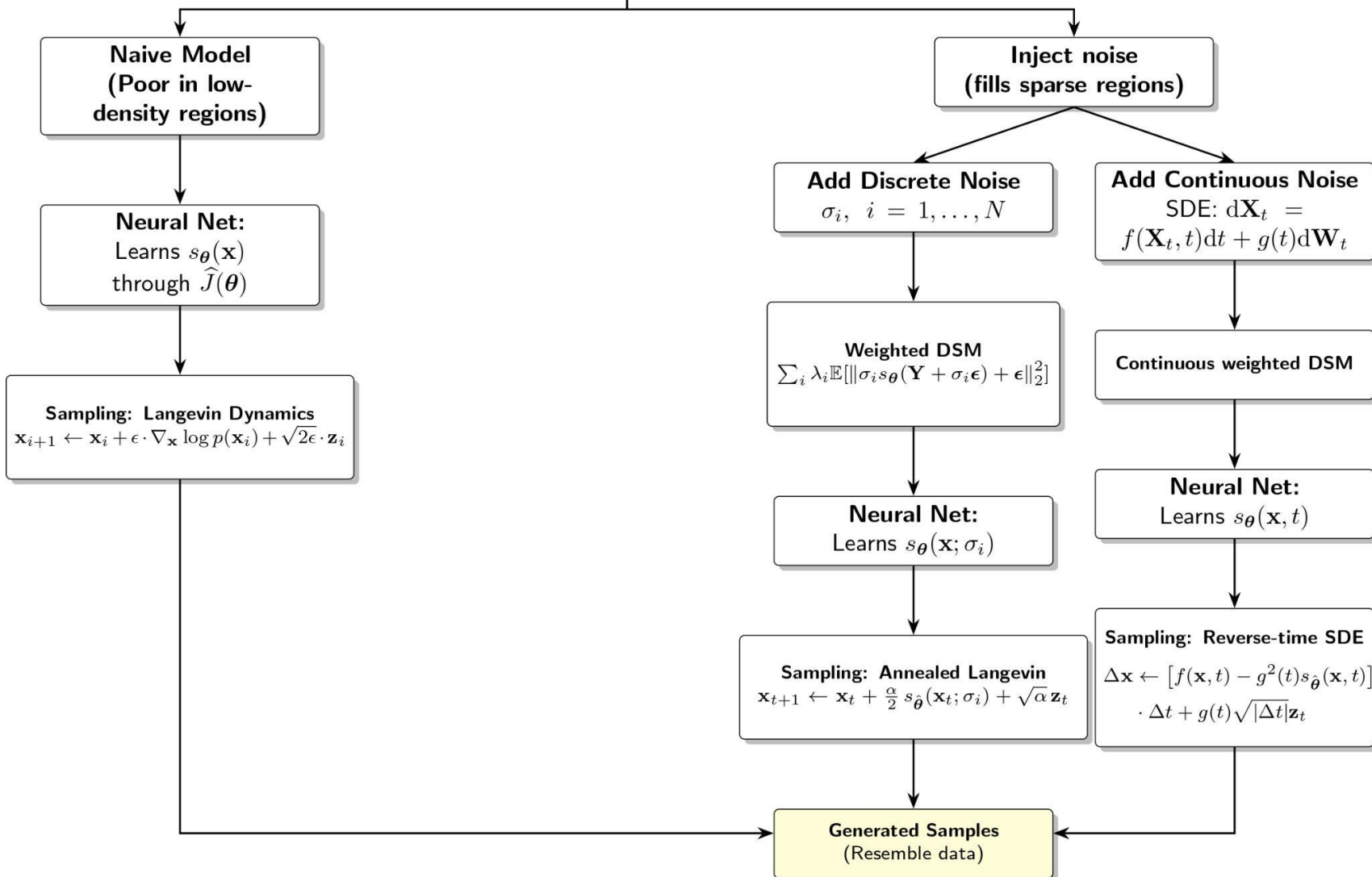
- ▶ The parametric model $\mathbf{s}_\theta$ can be trained by minimizing the *Fisher divergence*

$$L(\theta) := \frac{1}{2} \mathbb{E}_{\mathbf{x} \sim p_0} \left[\|\mathbf{s}_\theta(\mathbf{X}) - \mathbf{s}_0(\mathbf{X})\|_2^2 \right].$$

Score Function
For a density $p : \nabla_{\mathbf{x}} \log p(\mathbf{x})$

Chapter 2.2

Score Matching
$$\hat{J}(\theta) = \frac{1}{n} \sum_{i=1}^n \left[\nabla_{\mathbf{x}}^2 \log p_{\theta}(\mathbf{x}_i) + \frac{1}{2} \|s_{\theta}(\mathbf{x}_i)\|^2 \right]$$



Score matching

- ▶ A family of methods minimizes this objective without needing $\mathbf{s}_0$ is as *score matching*.
- ▶ **Aapo Hyvärinen**⁵ showed by applying integration by parts and some regularity conditions

$$L(\boldsymbol{\theta}) = \underbrace{\mathbb{E}_{\mathbf{x} \sim p_0} \left[\nabla_{\mathbf{x}}^2 \log p_{\boldsymbol{\theta}}(\mathbf{X}) + \frac{1}{2} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{X})\|^2 \right]}_{J(\boldsymbol{\theta})} + \underbrace{\frac{1}{2} \mathbb{E}_{\mathbf{x} \sim p_0} [\|\mathbf{s}_0(\mathbf{X})\|^2]}_C,$$

where $J(\boldsymbol{\theta})$ does not depend on the unknown true data score $\mathbf{s}_0$.

- ▶ The objective is then minimized by the sample version of $J(\boldsymbol{\theta})$

$$\widehat{J}(\boldsymbol{\theta}) := \frac{1}{n} \sum_{i=1}^n \left[\nabla_{\mathbf{x}}^2 \log p_{\boldsymbol{\theta}}(\mathbf{x}) + \frac{1}{2} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_i)\|^2 \right],$$

with n data points $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ in the dataset.

⁵ Hyvärinen (2005). “Estimation of non-normalized statistical models by score matching”. *Journal of Machine Learning Research*.

Example 1

- Consider $p_0 \equiv \mathcal{N}(0, \sigma_0^2)$ and $p \equiv \mathcal{N}(0, \theta)$.

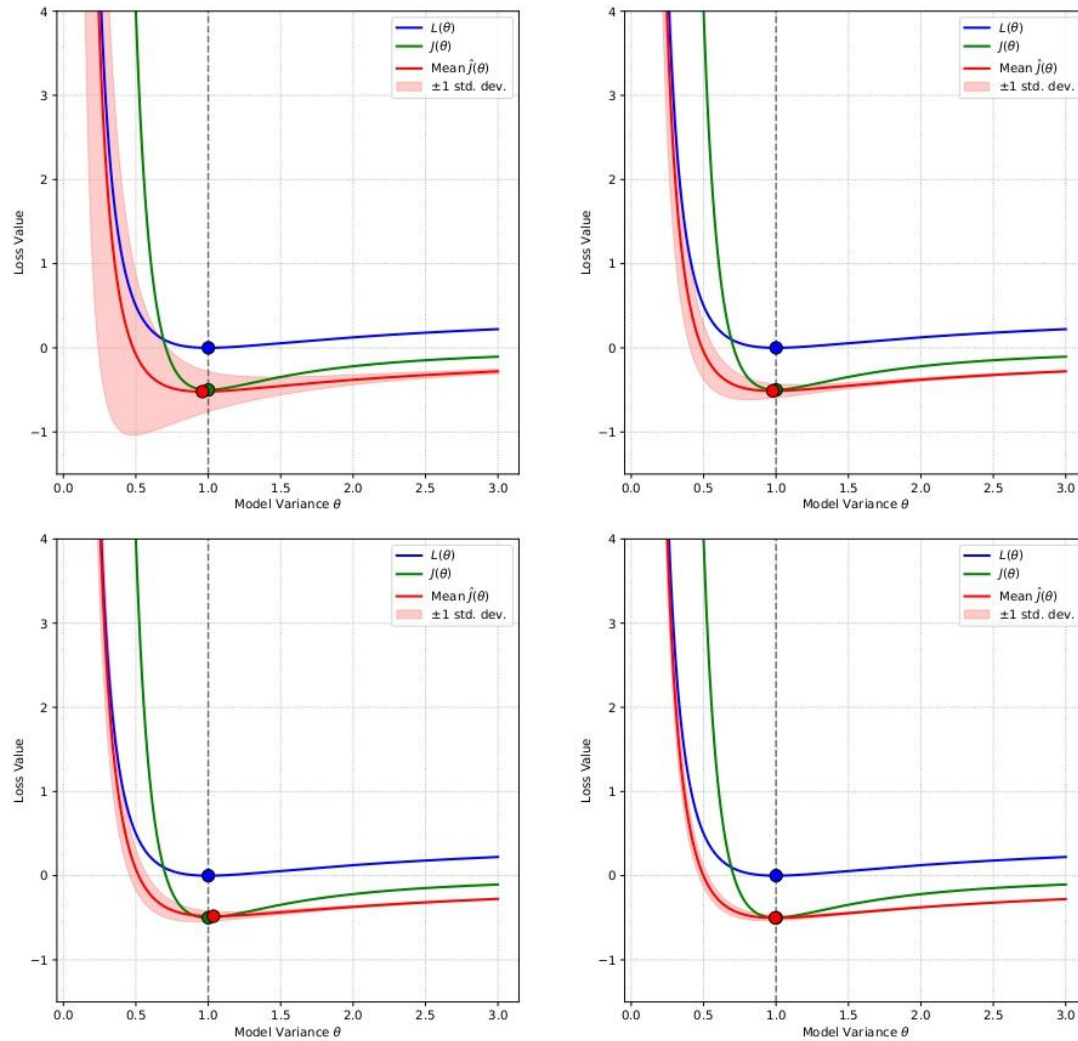
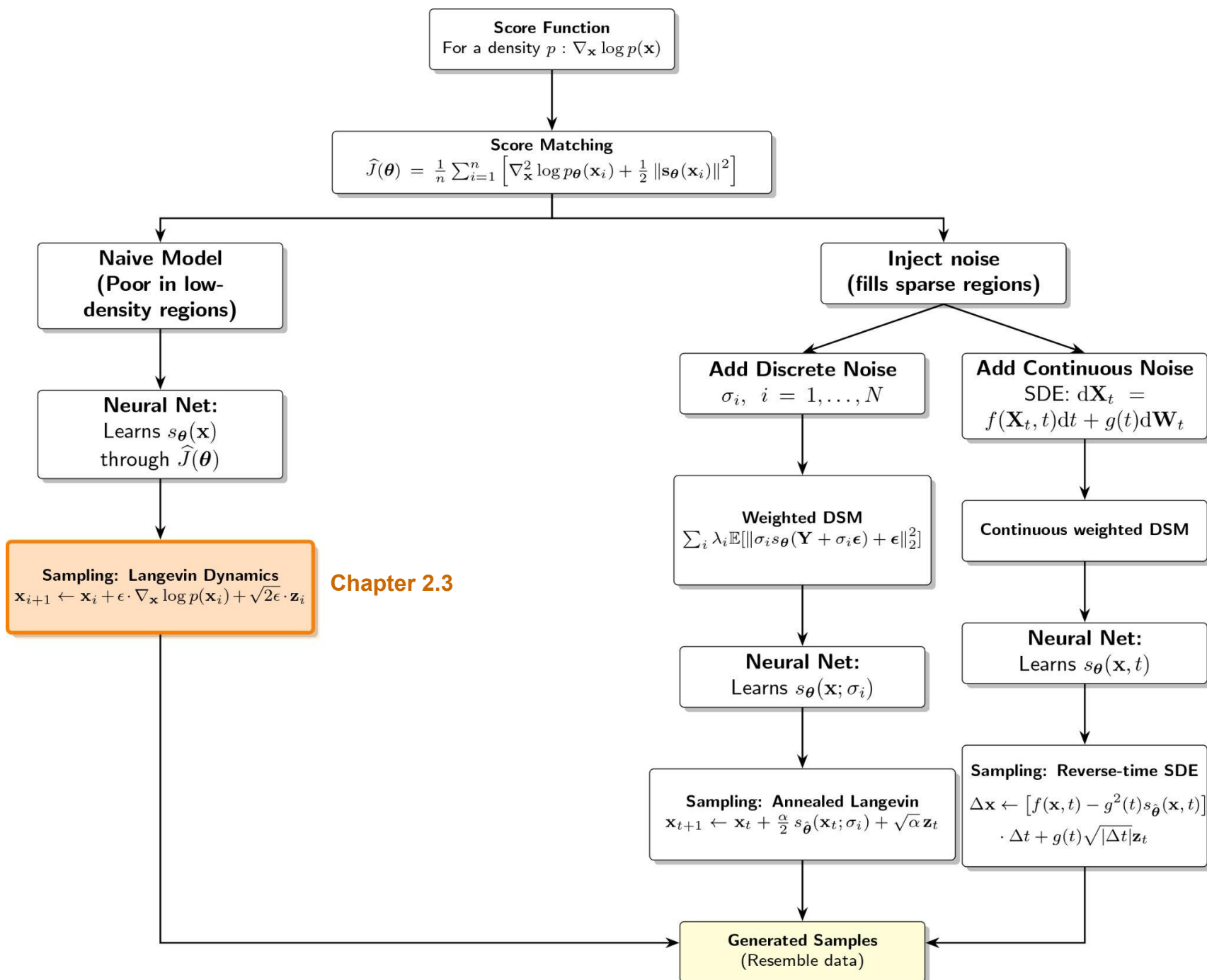


Figure: $L(\theta)$, $J(\theta)$, and $\hat{J}(\theta)$ vs. variance θ for 1D normal. Panels: $n = 10, 50, 100, 500$. Each: 50 runs; shaded regions show variability.



Langevin diffusions

- ▶ A continuous time process via the stochastic differential equation⁶

$$d\mathbf{X}_t = \nabla_{\mathbf{x}} \log p(\mathbf{X}_t) dt + \sqrt{2} d\mathbf{W}_t.$$

- ▶ Simulating this process can be done via Euler–Maruyama using discrete time steps⁷

$$\begin{aligned}\mathbf{X}_{t+\Delta t} &= \mathbf{X}_t + \int_t^{t+\Delta t} \nabla_{\mathbf{x}} \log p(\mathbf{X}_s) ds + \int_t^{t+\Delta t} \sqrt{2} d\mathbf{W}_s \\ &\approx \mathbf{X}_t + \nabla_{\mathbf{x}} \log p(\mathbf{X}_t) (\Delta t) + \sqrt{2}(\sqrt{\Delta t} \mathbf{Z}).\end{aligned}$$

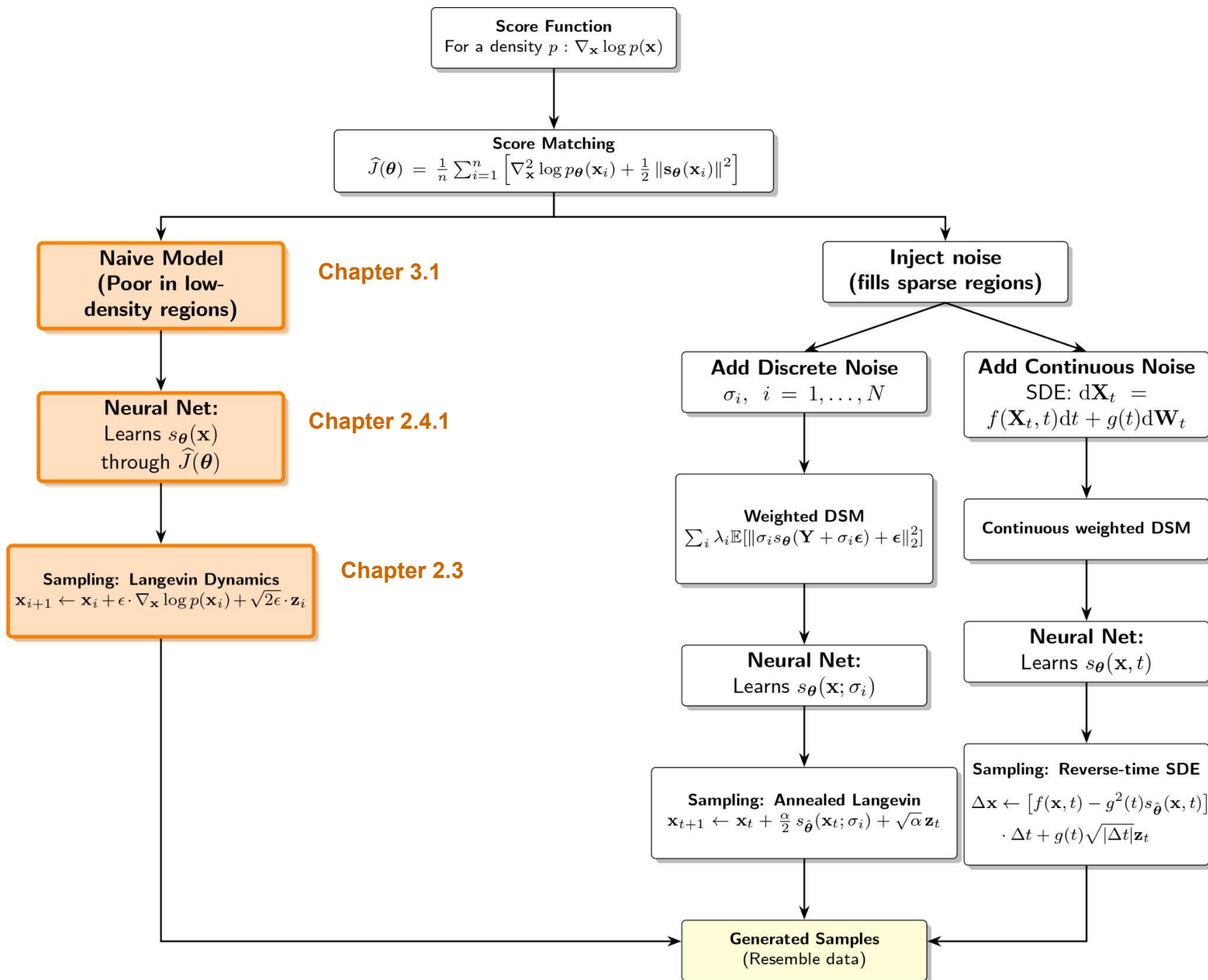
- ▶ The final discretized update rule then is

$$\mathbf{x}_{i+1} \leftarrow \mathbf{x}_i + \nabla_{\mathbf{x}} \log p(\mathbf{x}) \cdot \epsilon + \sqrt{2\epsilon} \cdot \mathbf{z}_i,$$

where $i = 0, 1, \dots, N$, and each $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is drawn independently.

⁶ Roberts et al. (1996). “Exponential convergence of Langevin distributions and their discrete approximations”. *Bernoulli*.

⁷ Higham (2001). “An algorithmic introduction to numerical simulation of stochastic differential equations”. *SIAM Review*.



Naive model + Example 2

- The Fisher divergence is weighted by the probability density p_0 itself⁸

$$\mathbb{E}_{\mathbf{x} \sim p_0} [\|\nabla_{\mathbf{x}} \log p_0(\mathbf{x}) - s_{\theta}(\mathbf{x})\|_2^2] = \int \|\nabla_{\mathbf{x}} \log p_0(\mathbf{x}) - s_{\theta}(\mathbf{x})\|_2^2 p_0(\mathbf{x}) d\mathbf{x}.$$

- **Example 2.** We define a 2D oval-shaped distribution and train an MLP network to learn its score function using explicit score matching $\hat{J}(\theta)$.

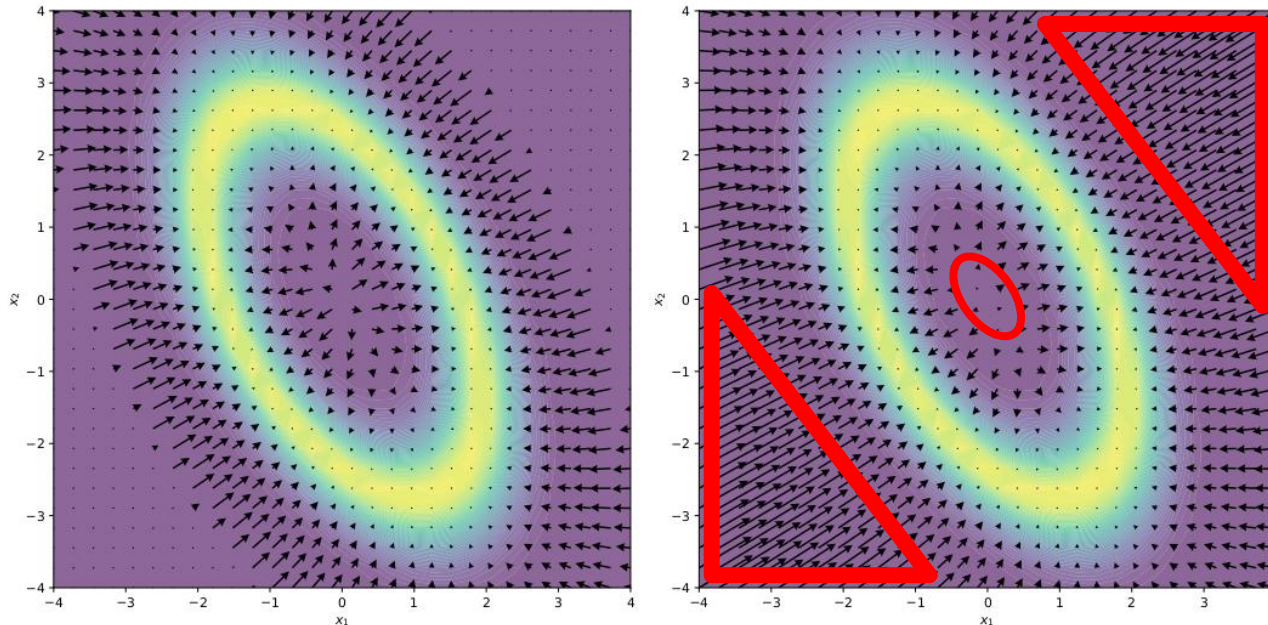


Figure: **Left:** Ground-truth score vectors **Right:** Neural net predictions. Background shows the true density.

⁸ Song et al. (2019). “Generative Modeling by Estimating Gradients of the Data Distribution”. In: *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Example 2. Continued

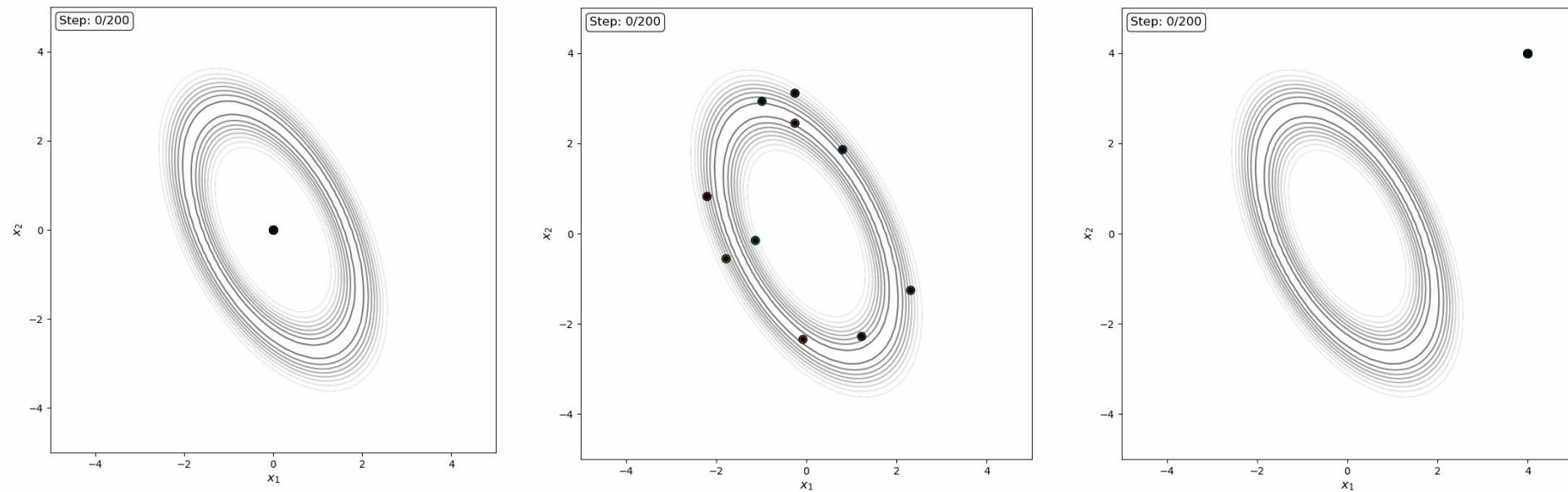
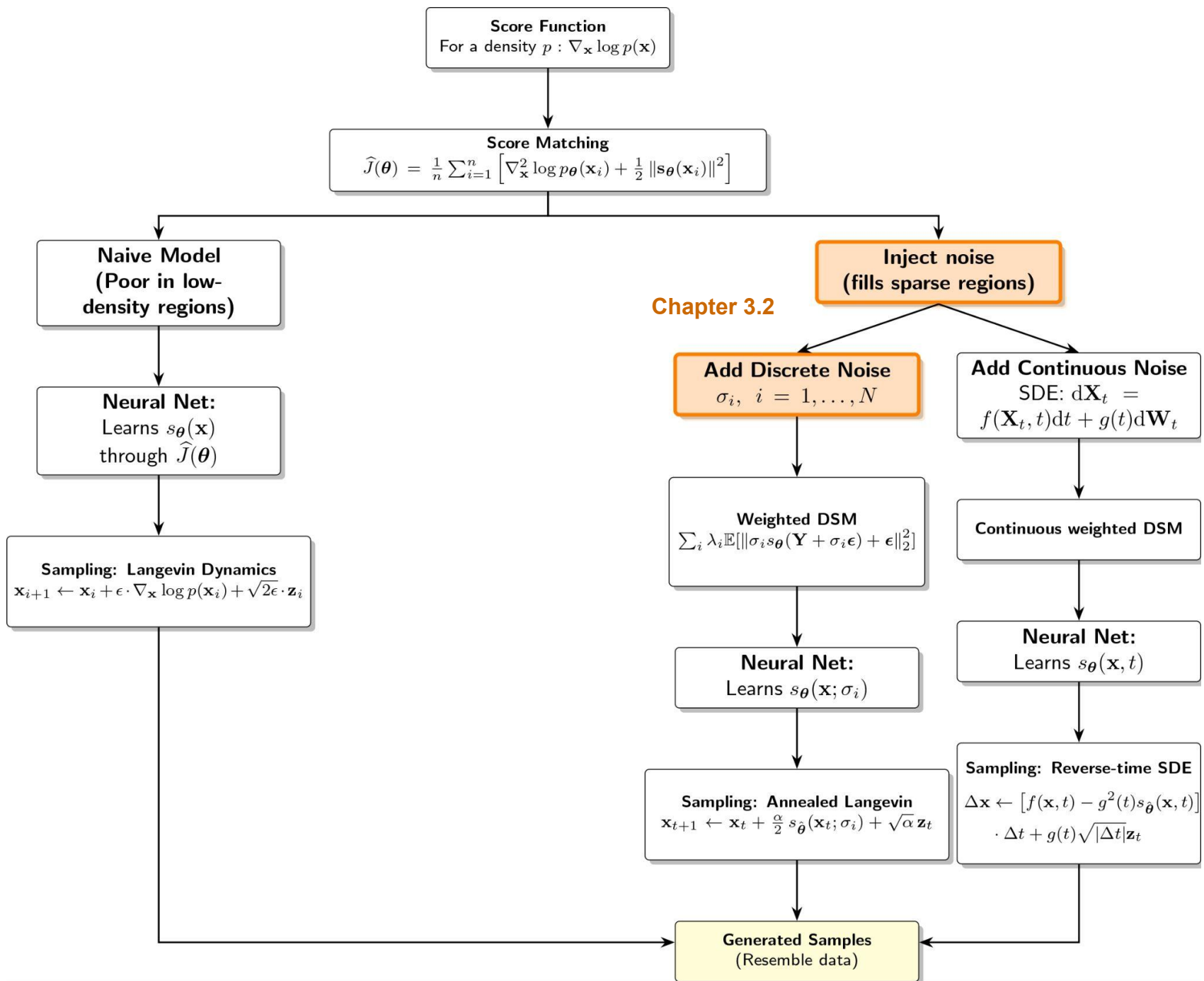


Figure: Langevin Markov chains from three different initializations (trajectories). Density contours show the true density.



Adding noise

- ▶ Adding noise to the original samples can address the issues in low-density regions⁹.
- ▶ The new density is a convolution with a Gaussian kernel

$$p_0(\mathbf{x}; \sigma) := (p_0 * \phi_{\sigma^2 \mathbf{I}})(\mathbf{x}) = \int_{\mathbb{R}^d} \phi_{\sigma^2 \mathbf{I}}(\mathbf{y}) p_0(\mathbf{x} - \mathbf{y}) d\mathbf{y} = \int_{\mathbb{R}^d} \phi_{\sigma^2 \mathbf{I}}(\mathbf{x} - \mathbf{y}) p_0(\mathbf{y}) d\mathbf{y}.$$

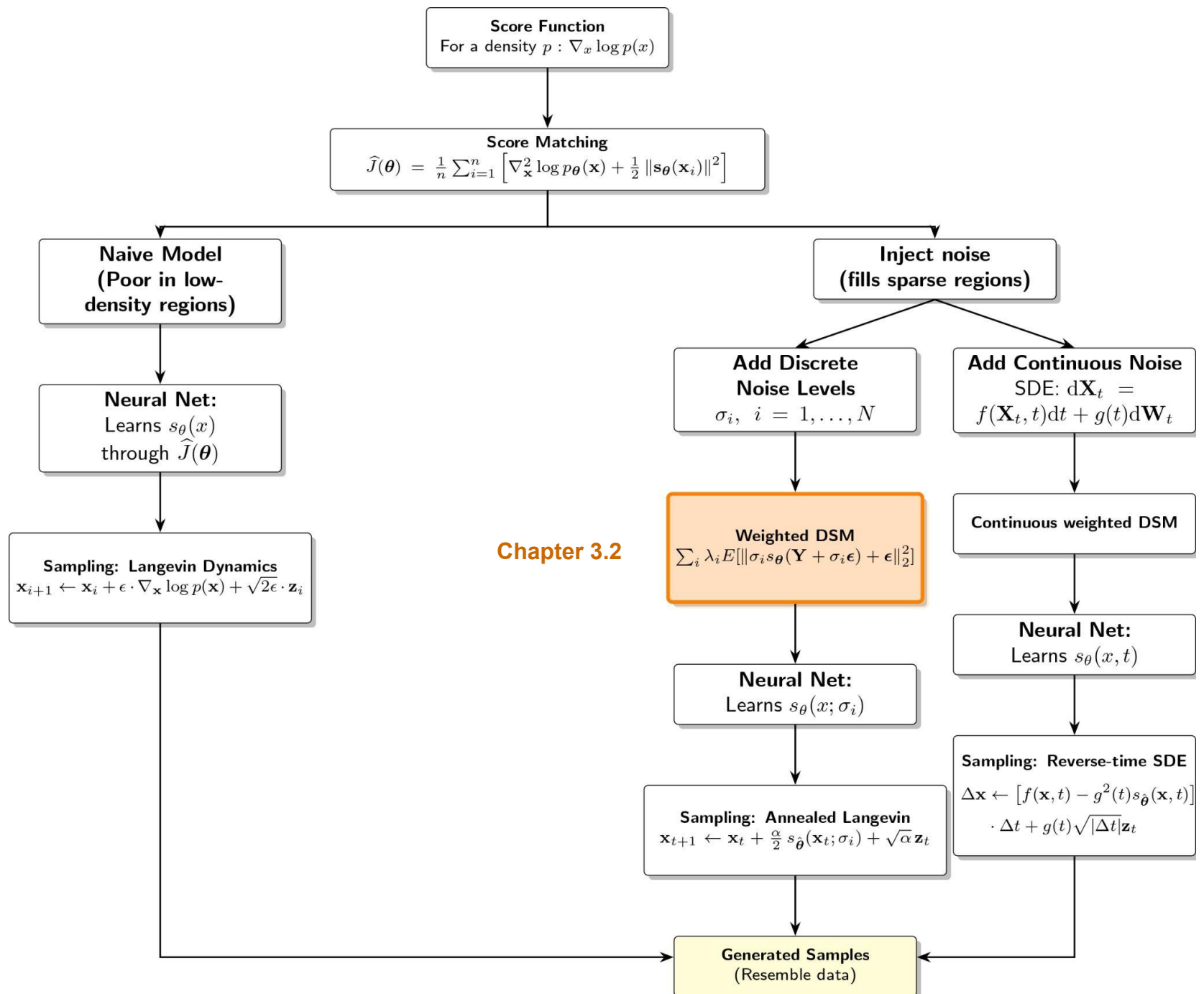
- ▶ Multiple noise levels are required to perturb the original data:

$$s_{\hat{\theta}}(\mathbf{x}; \sigma_i) \approx \nabla_{\mathbf{x}} \log p_0(\mathbf{x}; \sigma_i), \quad \forall i = 1, \dots, L.$$

- ▶ **Loss function:** a weighted sum of Fisher divergences for all noise scales:

$$\sum_{i=1}^L \lambda_i \mathbb{E}_{\mathbf{X} \sim p_0(\cdot; \sigma_i)} \left[\|s_{\theta}(\mathbf{X}; \sigma_i) - \nabla_{\mathbf{x}} \log p_0(\mathbf{X}; \sigma_i)\|_2^2 \right].$$

⁹ Song et al. (2020). “Score-based generative modeling through stochastic differential equations”. [arXiv e-prints](#).



Denoising score matching

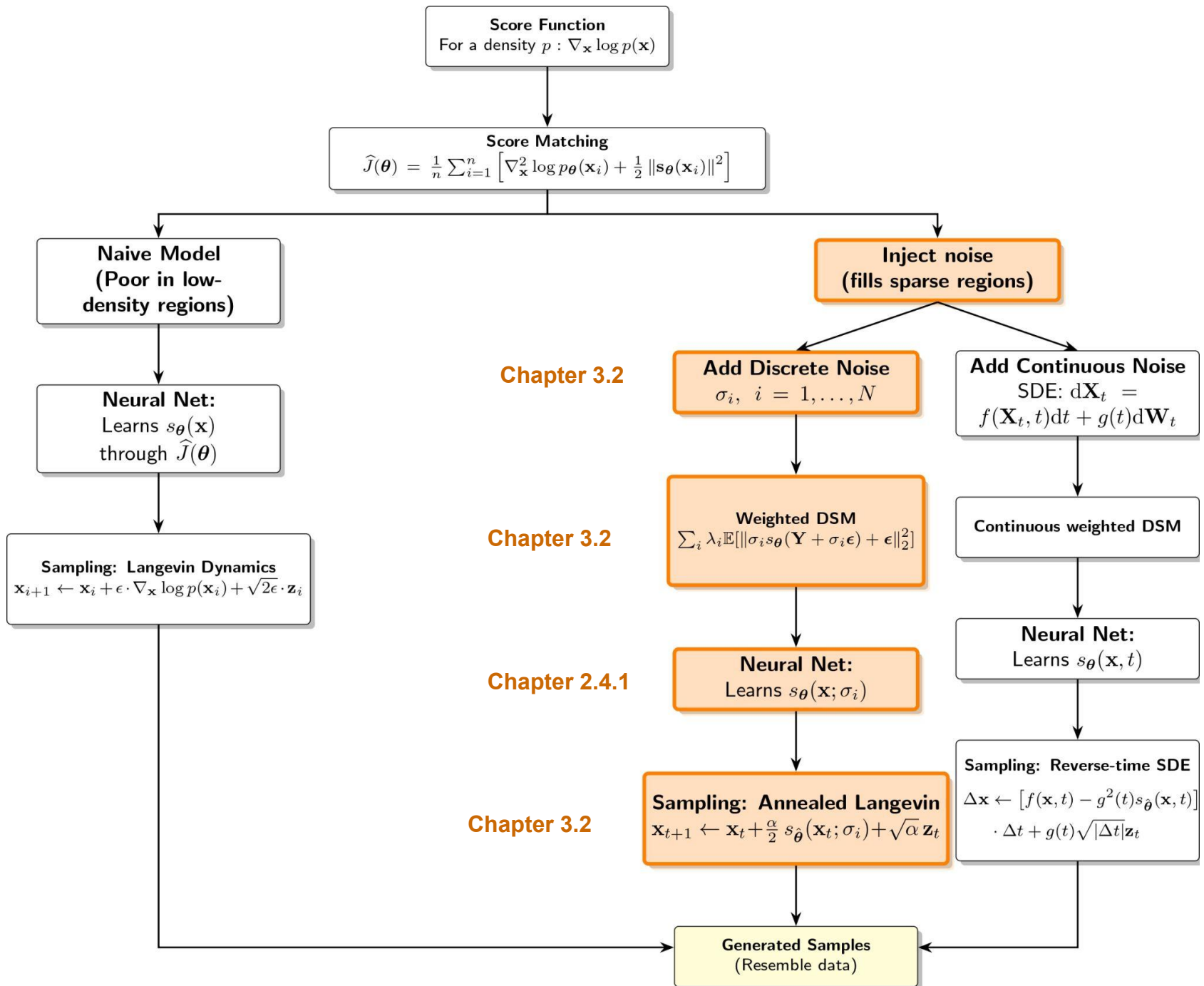
- ▶ **Pascal Vincent**¹⁰ showed that, by adding Gaussian noise to the data, the Fisher divergence objective can be rewritten as a practical *denoising score matching* (DSM) loss

$$\tilde{J}_{\text{DSM, weighted}}(\theta) = \sum_{i=1}^L \lambda_i \cdot \mathbb{E}_{\mathbf{Y} \sim p_0, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\left\| s_{\theta}(\mathbf{Y} + \sigma_i \epsilon) + \frac{\epsilon}{\sigma_i} \right\|_2^2 \right].$$

- ▶ The weighting function λ_i is often set to σ_i^2

$$\tilde{J}_{\text{DSM, weighted}}(\theta) = \sum_{i=1}^L \mathbb{E}_{\mathbf{Y} \sim p_0, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\left\| \sigma_i s_{\theta}(\mathbf{Y} + \sigma_i \epsilon) + \epsilon \right\|_2^2 \right].$$

¹⁰ Vincent (2011). “A Connection Between Score Matching and Denoising Autoencoders”. *Neural Computation*.



Annealed Langevin sampling + Example 3

- Start from noise (highest σ), then denoise step-by-step using model scores via Langevin dynamics

$$\mathbf{x}_{t+1} \leftarrow \mathbf{x}_t + \frac{\alpha}{2} s_{\hat{\theta}}(\mathbf{x}_t; \sigma_i) + \sqrt{\alpha} \mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

- **Example 3.** As in Example 2, but each data point gets different Gaussian noise and the model learns the score function for each noise scale using denoising score matching.

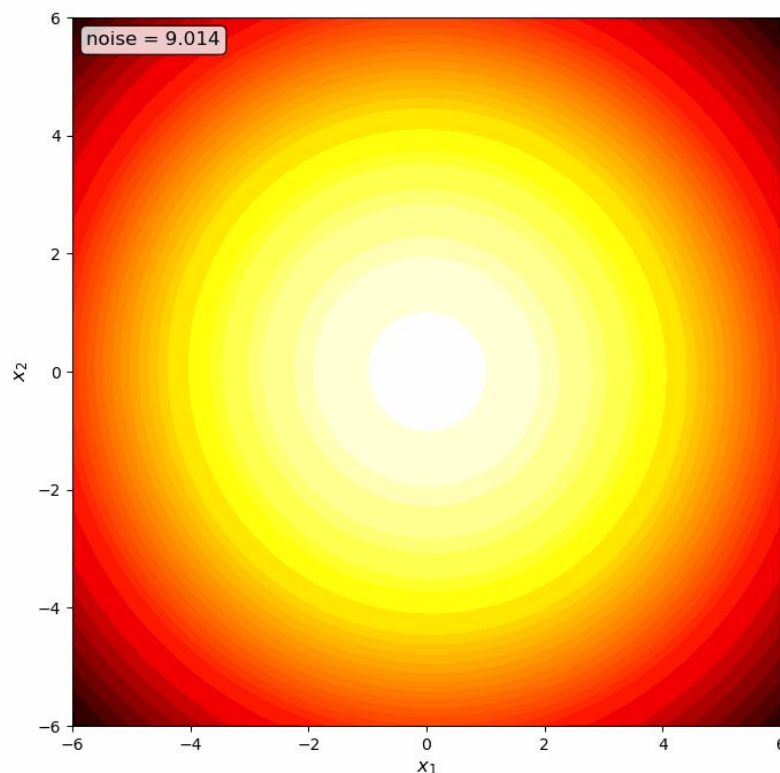


Figure: Animation of perturbed densities as noise transitions from high to low.

Example 3. Continued

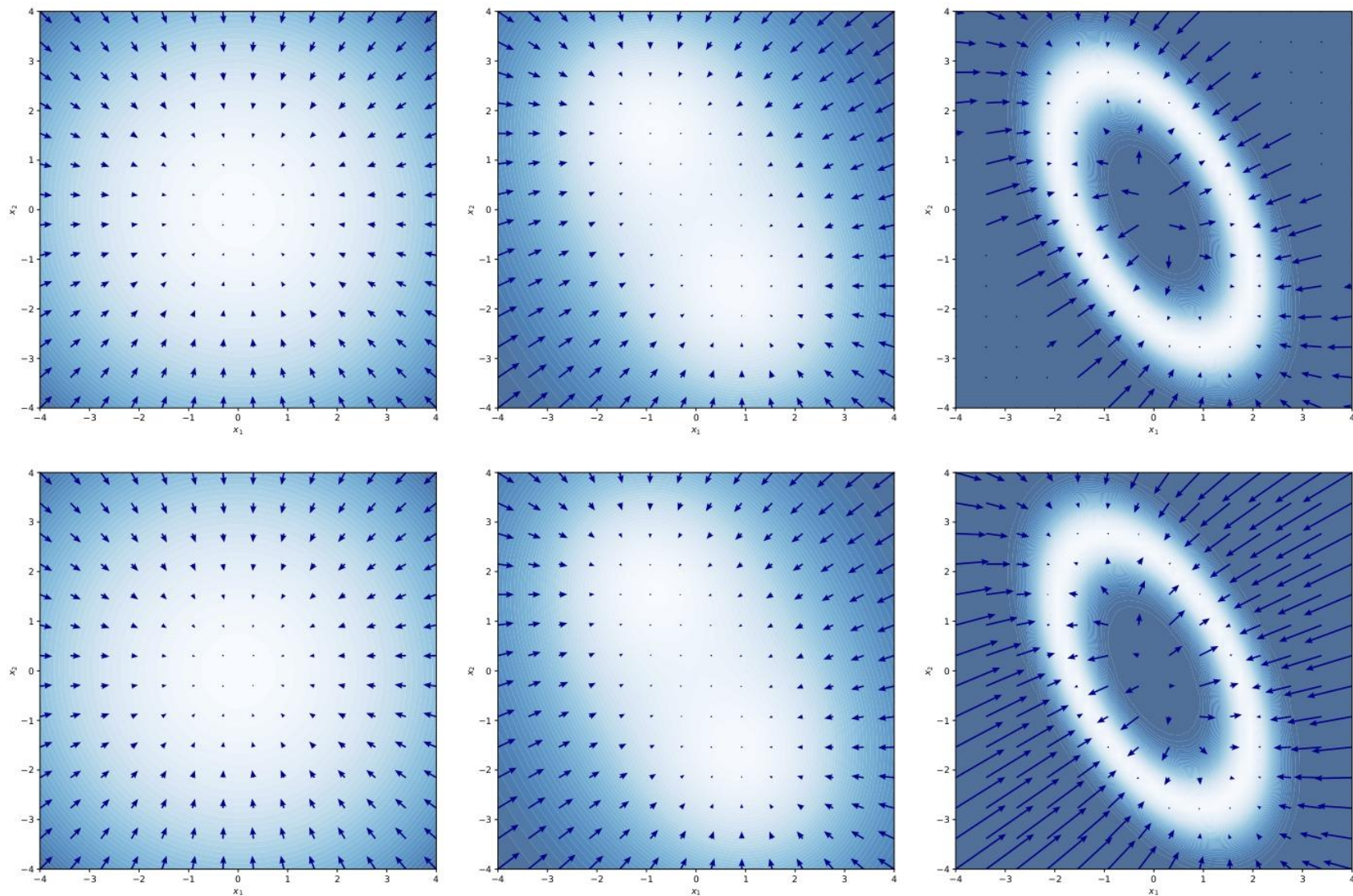


Figure: True (top) vs. learned (bottom) score fields at high, medium, and low noise levels.

Example 3. Continued

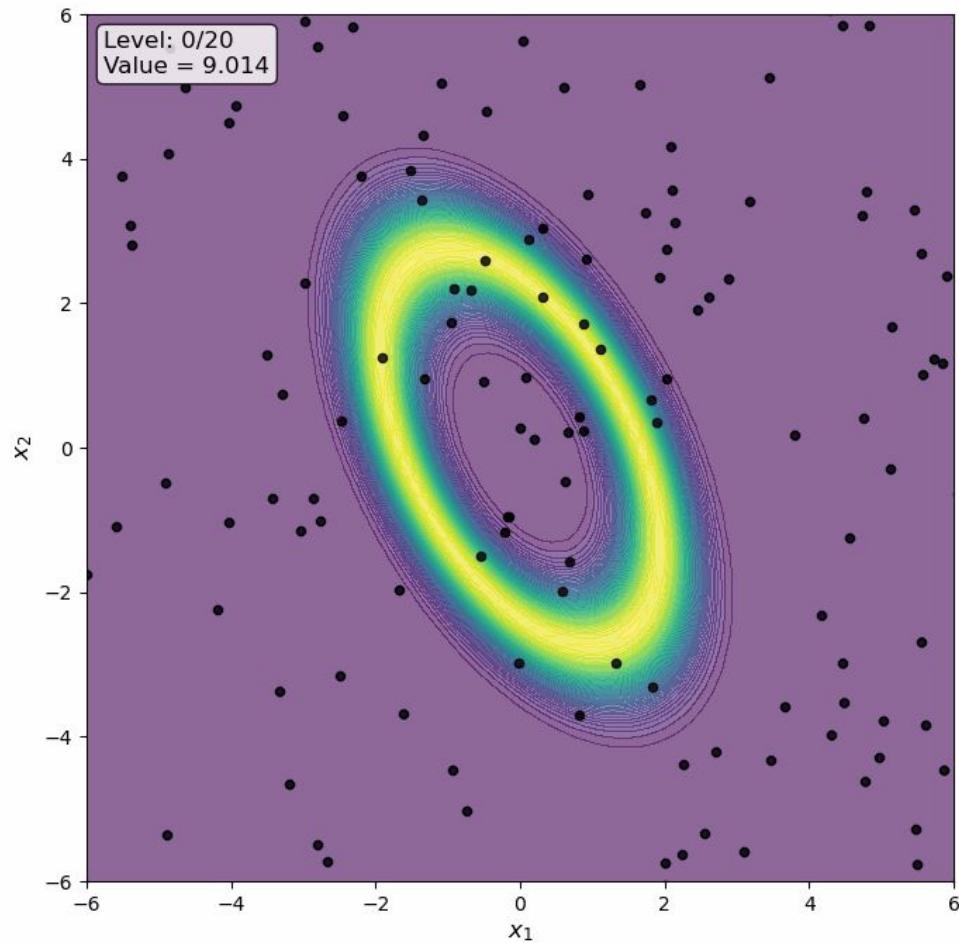
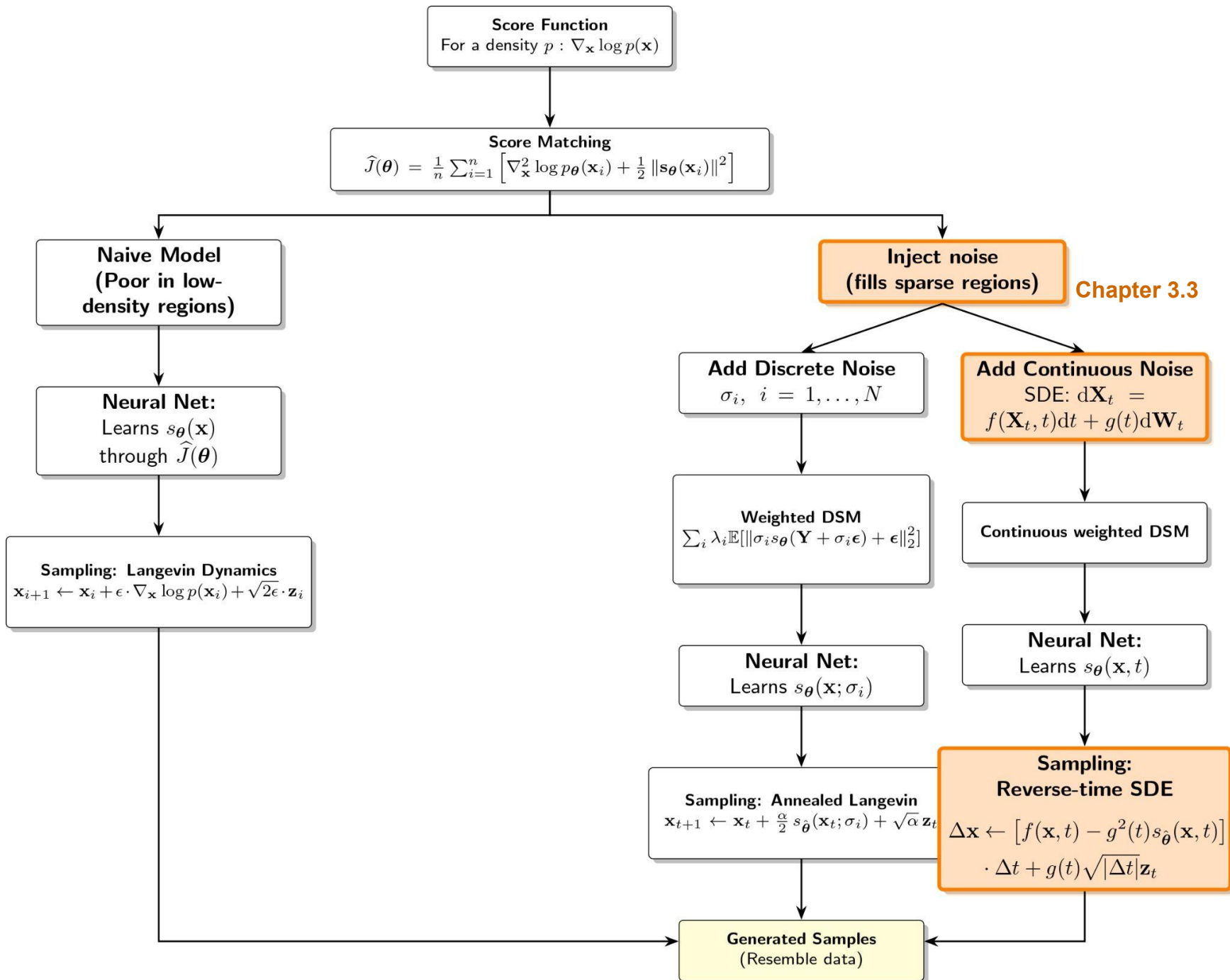


Figure: Animation of sample evolution during denoising, illustrating the transition from high to low noise. The animation visualizes the gradual recovery of the true data density as noise is reduced.



Score SDEs

- ▶ Adding noise continuously makes the process a continuous-time stochastic process defined by an SDE¹⁰

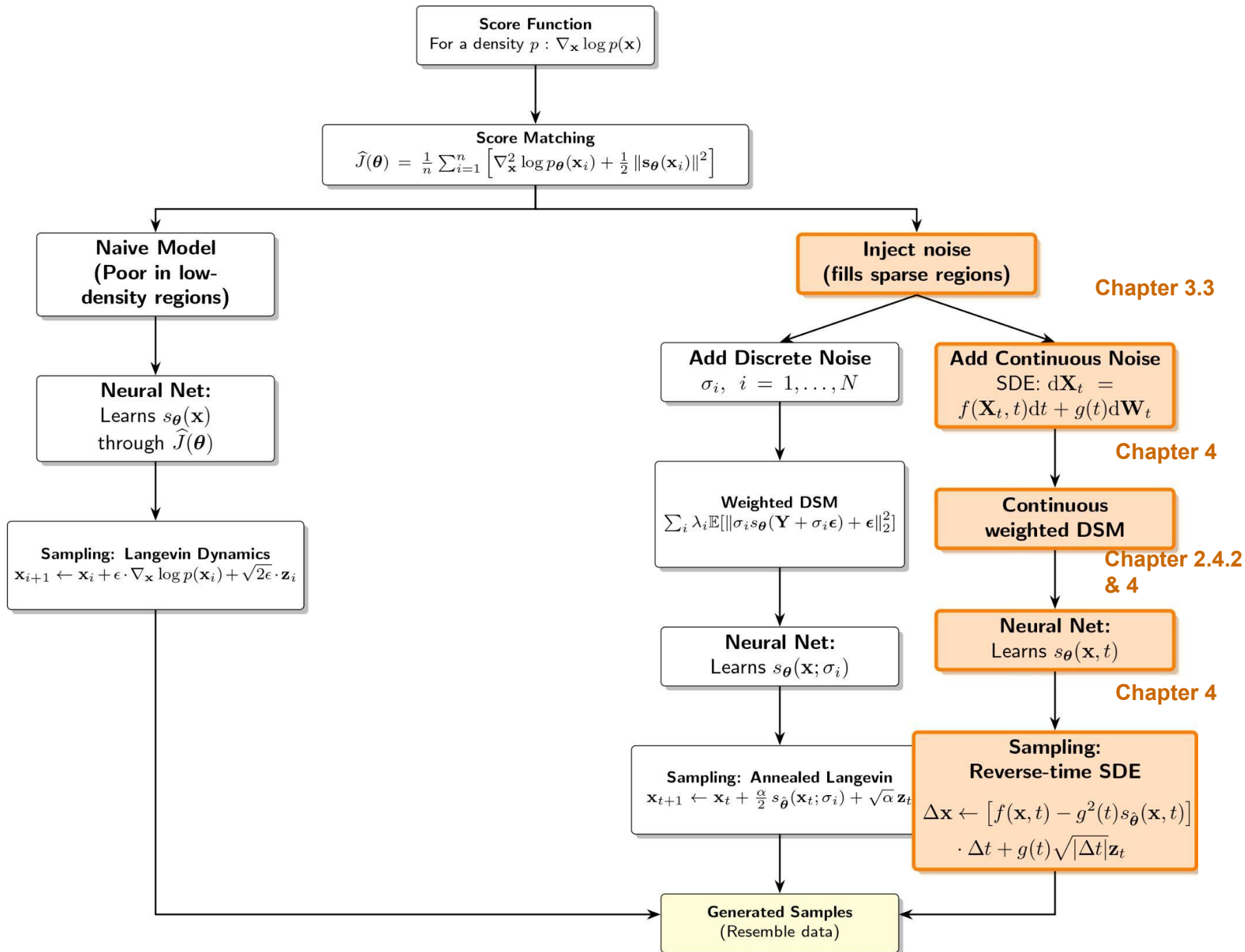
$$d\mathbf{X}_t = f(\mathbf{X}_t, t) dt + g(t) d\mathbf{W}_t.$$

- ▶ This SDE is not restricted to a specific form. Discrete noise is a special case with zero drift and exponentially growing diffusion, adding only noise.
- ▶ There exists a reverse SDE associated with any forward SDE, which helps with generating new samples by transforming pure noise to data¹¹

$$d\mathbf{X}_t = \left[f(\mathbf{X}_t, t) - g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{X}_t) \right] dt + g(t) d\mathbf{W}_t.$$

¹⁰ Song et al. (2020). “Score-based generative modeling through stochastic differential equations”. *arXiv e-prints*.

¹¹ Anderson (1982). “Reverse-time diffusion equation models”. *Stochastic Processes and their Applications*.



Experiment

- ▶ A U-Net architecture to process 28×28 gray-scale images (main implementation adopted from Yang Song¹²).
- ▶ A VE-SDE is chosen as the process to add noise to clean data

$$d\mathbf{X} = \sigma^t d\mathbf{W}, \quad t \in [0, 1].$$

- ▶ Similar to Example 5, the loss function is defined as the weighted denoising score matching loss.
- ▶ The corresponding reverse SDE is simulated using Euler–Maruyama

$$\mathbf{x}_{t-\Delta t} \leftarrow \mathbf{x}_t + \sigma^{2t} s_{\hat{\theta}}(\mathbf{x}_t, t) \Delta t + \sigma^t \sqrt{\Delta t} \mathbf{Z}_t, \quad \mathbf{Z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

¹²Song (2021). *Generative modeling by estimating gradients of the data distribution*. [Blog post](https://yang-song.net/blog/2021/score/). URL: <https://yang-song.net/blog/2021/score/>.

“Quick, Draw!” dataset

- ▶ A dataset collected from millions of users drawing simple sketches online as part of a game developed by Google.¹³
- ▶ **Tree:** 144,721. **Car:** 182,764. **Octopus:** 150,152.

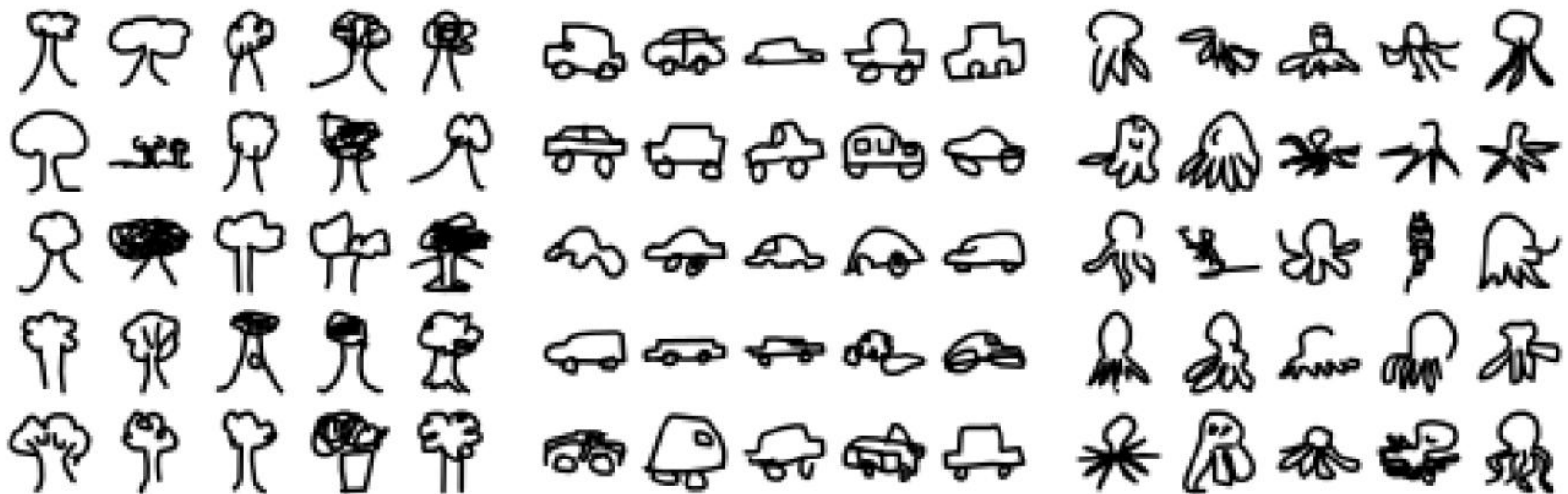


Figure: Several samples from each dataset: **Left:** Tree, **Middle:** Car, and **Right:** Octopus.

¹³ Ha et al. (2018). “A Neural Representation of Sketch Drawings”. In: *Proceedings of the International Conference on Learning Representations (ICLR)*.

Trees

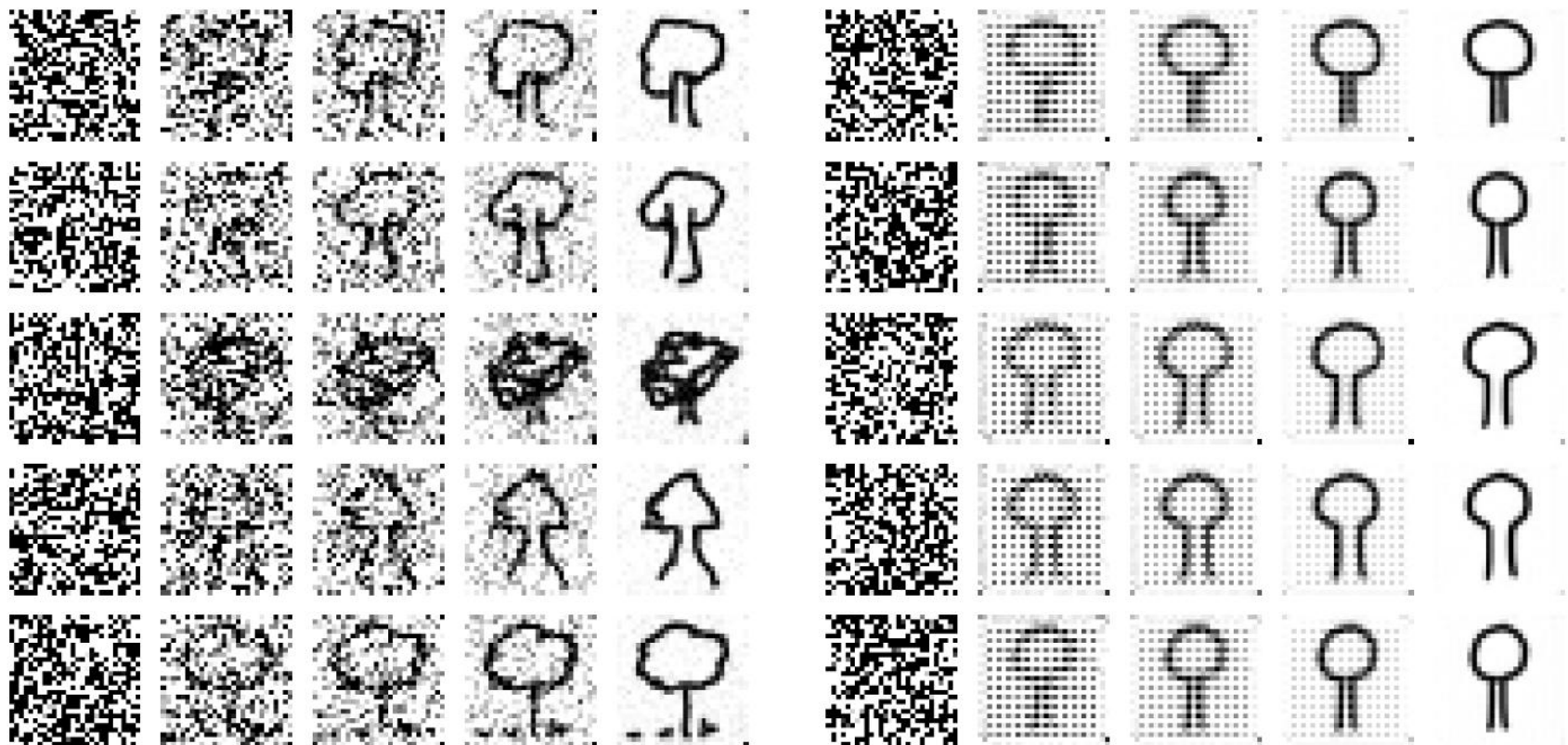


Figure: Denoising on the tree dataset: with Gaussian noise (left) vs. without (right). Each row shows an independent sample over 1,000 iterations.

Cars

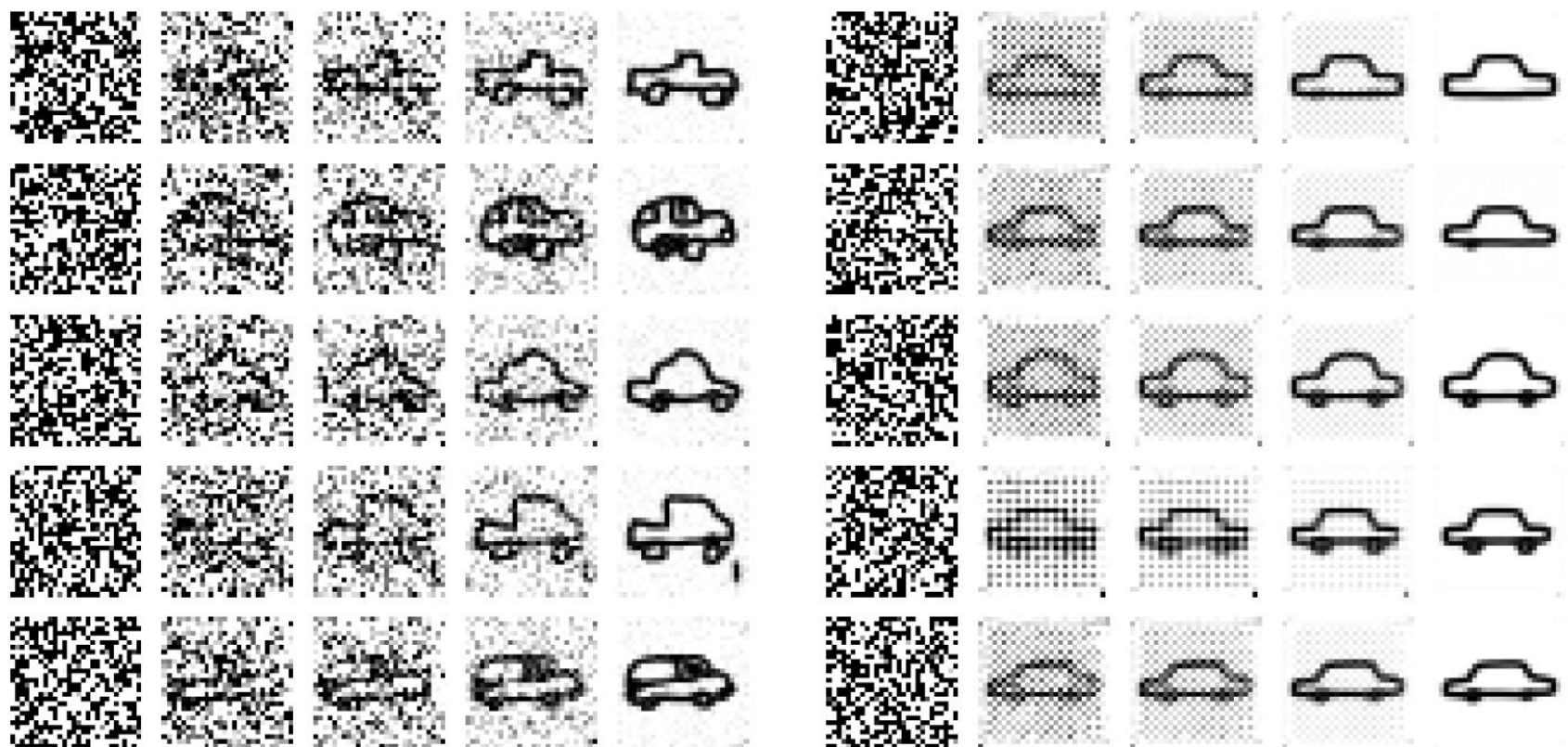


Figure: Denoising on the car dataset: with Gaussian noise (left) vs. without (right). Each row shows an independent sample over 1,000 iterations.

Octopuses

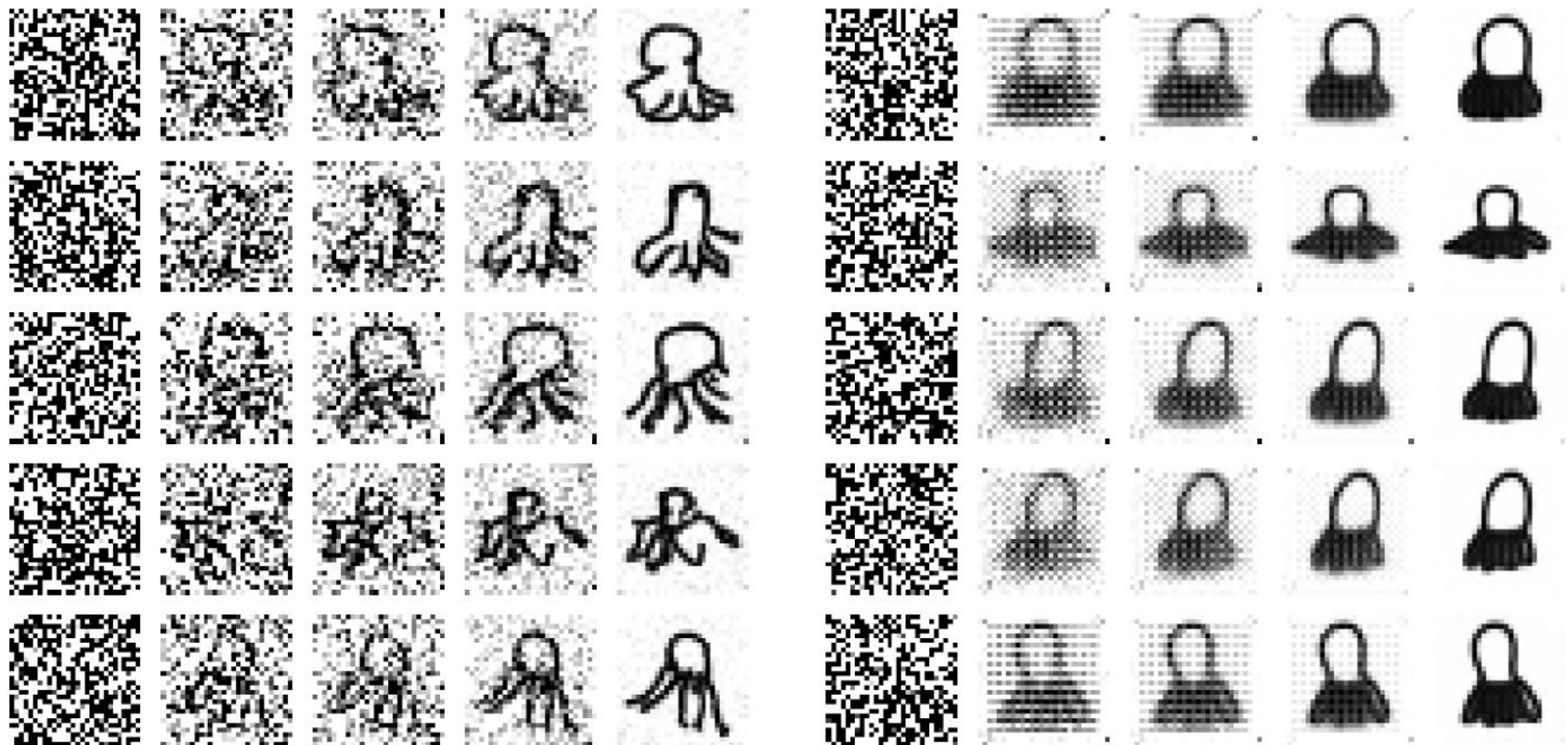


Figure: Denoising on the octopus dataset: with Gaussian noise (left) vs. without (right). Each row shows an independent sample over 1,000 iterations.

Conclusion

- ▶ **Clarified the fundamentals** of score-based diffusion models and their core mathematical elements.
- ▶ Explained **key building blocks**: score function, score matching, Langevin dynamics, neural networks, and SDEs.
- ▶ Demonstrated limitations of standard score matching and how **injecting noise** improves learning.
- ▶ Connected theory and practice through original intuitive examples and visualizations.